SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

Dr

QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	5
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	5
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	5
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	5
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	5
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	5
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	5

8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	**5**
9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	**5**
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	**5**

Code of Conduct:

I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Judy Allen J

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated

Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Q1. Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?
Answer:
Query processing converts an SQL query into an efficient execution plan.
Steps:
1. **Parsing and Translation:** Checks SQL syntax and converts the query into an internal representation such as a relational algebra expression.
2. **Optimization:** Generates alternative execution plans.
3. **Plan Selection:** Estimates the cost of each plan based on disk I/O, CPU time, indexes, table size, and join methods.
4. **Execution:** The lowest-cost plan is selected and executed.

Example: If an index exists on Student_ID, the optimizer may use an index scan instead of scanning the entire Student table.

Q2. What is the difference between heuristic optimization and cost-based optimization? When would you use each?
Answer:

Heuristic Optimization	Cost-Based Optimization
Uses predefined optimization rules.	Evaluates multiple execution plans.
Does not calculate the exact cost of plans.	Estimates CPU, I/O and memory costs.
Faster optimization process.	More optimization time is required.
Example: Perform selection before join.	Example: Choose index scan over full table scan based on cost.

Usage: Heuristic optimization is suitable for simple queries, while cost-based optimization is preferred for complex queries with joins and large tables.

Q3. How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.
Answer:
Database connectivity allows an application to communicate with a DBMS.
Implementation steps:
1. Install/load the required database driver.
2. Establish a connection using the database URL and credentials.
3. Create and execute SQL statements.
4. Retrieve and process the results.
5. Close the connection and other resources.

JDBC vs ODBC:

- **JDBC (Java Database Connectivity):** Designed mainly for Java applications and uses JDBC drivers.
- **ODBC (Open Database Connectivity):** Language-independent standard that can connect applications written in different languages to databases.

Example: A Java application can use JDBC to connect with a MySQL database.

Q4. What are ACID properties? Give a real-world example where each property is critical.
Answer:
ACID properties ensure reliable database transactions.

1. **Atomicity:** A transaction is completed fully or not at all.
   Example: During a bank transfer, both debit and credit must occur.
2. **Consistency:** A transaction maintains database rules and constraints.
   Example: An account cannot violate predefined balance constraints.
3. **Isolation:** Concurrent transactions should not interfere with each other.
   Example: Two users booking the same seat should not both receive it.
4. **Durability:** Committed changes remain permanent even after system failure.
   Example: Once payment is confirmed, the transaction must remain recorded after a server restart.

Q5. Explain dirty read, lost update, and unrepeatable read. How does 2PL solve them?
Answer:
These problems occur when multiple transactions execute concurrently.

- **Dirty Read:** A transaction reads data modified by another transaction before it is committed.
- **Lost Update:** One transaction's update is overwritten by another transaction.
- **Unrepeatable Read:** Reading the same record twice gives different values because another transaction modified it.

Two-Phase Locking (2PL) controls concurrent access using locks.
It has two phases:

1. **Growing Phase:** Locks can be acquired but not released.
2. **Shrinking Phase:** Locks can be released but no new locks can be acquired.

Thus, 2PL ensures conflict serializability and prevents conflicting operations from executing simultaneously. Strict forms of 2PL also prevent dirty reads.

Q6. What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.

Answer:

A **deadlock** occurs when two or more transactions wait indefinitely for resources locked by each other.

Example:

Transaction T1 holds resource A and waits for B, while T2 holds B and waits for A.

Deadlock Detection:

A DBMS creates a **Wait-For Graph**:

- Transactions are represented as nodes.
- A directed edge T1 → T2 means T1 is waiting for T2.
- If the graph contains a **cycle**, a deadlock exists.
- The DBMS selects a victim transaction and rolls it back.

Prevention Strategies:

1. **Wait-Die:** Older transactions wait; younger transactions are rolled back.
2. **Wound-Wait:** Older transactions force younger transactions to roll back.

Q7. Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?

Answer:

Immediate Update	Deferred Update
Database can be updated before commit.	Database is updated only after commit.
Requires **UNDO and REDO** operations.	Mainly requires **REDO**.
More complex recovery.	Simpler recovery mechanism.
Suitable when early updates are required.	Suitable when transactions can wait until commit.

Preference: Immediate Update is useful for systems requiring frequent updates and greater flexibility. Deferred Update is preferred when simpler and safer recovery is required.

Q8. Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?
Answer:
Shadow Paging is a recovery technique that maintains two page tables:

- **Shadow Page Table:** Contains the original database pages and remains unchanged during the transaction.
- **Current Page Table:** Contains newly modified pages.

When the transaction commits, the current page table becomes the new page table. If the transaction fails, the system uses the unchanged shadow page table.

Advantages:

- No UNDO/REDO log is required.
- Recovery is fast.
- Simple recovery mechanism.

Limitations:

- Page-table copying creates overhead.
- Can cause data fragmentation.
- Not efficient for large databases or highly concurrent transactions.

Therefore, log-based recovery is generally more suitable for large multi-user systems.

Q9. What is a serializability schedule? How do you test for conflict serializability using a precedence graph?
Answer:
A serial schedule executes transactions one after another. A serializable schedule allows concurrent execution but produces a result equivalent to some serial schedule.
To test conflict serializability, a precedence graph is created.

Steps:

1. Create one node for each transaction.
2. Identify conflicting operations on the same data item.
3. If an operation of T1 occurs before a conflicting operation of T2, add an edge T1 → T2.
4. Check the graph for cycles.

Result:

- No cycle → Conflict Serializable
- Cycle present → Not Conflict Serializable

Thus, the precedence graph determines whether concurrent transactions can safely behave like a serial execution.

Q10. Describe Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?

Answer:

1. Basic 2PL:

Transactions have a growing phase for acquiring locks and a shrinking phase for releasing locks. It ensures conflict serializability but may cause cascading rollback.

2. Strict 2PL:

All exclusive/write locks are held until the transaction commits or aborts. This prevents dirty reads and cascading rollback.

3. Rigorous 2PL:

Both shared/read locks and exclusive/write locks are held until commit or abort. It provides stronger isolation but reduces concurrency.

Most commonly used:

Strict 2PL is widely preferred because it provides a good balance between serializability, recoverability, and concurrency, while also preventing cascading rollbacks.